

# Individual Claim Reserving

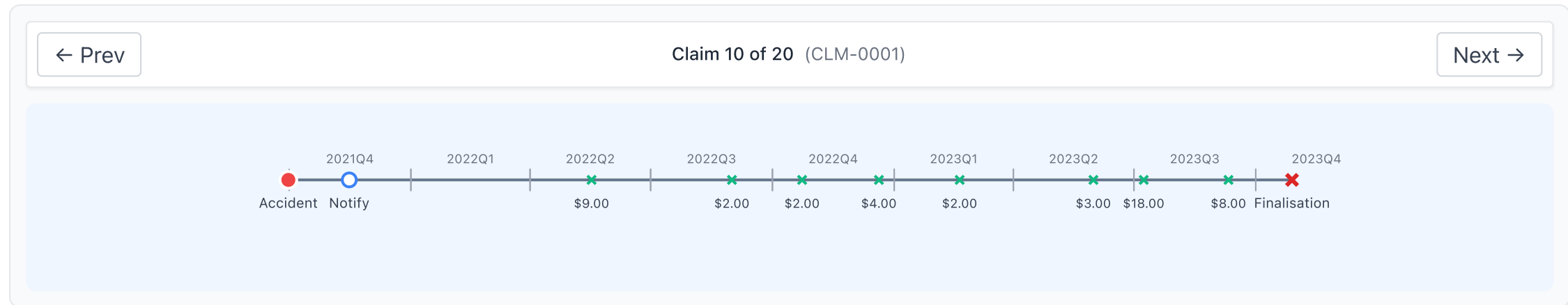
ACTL3143 & ACTL5111 Deep Learning for Actuaries  
Patrick Laub

# Lecture Outline

- **Introduction**
- Insurance Claim Payments and Case Estimates
- Benchmark: Aggregate Reserving
- Individual Claim Modelling Choices
- Preprocessing One Claim
- Splitting Claims by Finalisation Quarter
- Splitting Claims by Notification Quarter (Advanced)
- Case Study: Synthetic Claims
- Fitting the Models
- Individual-level Metrics
- Portfolio-level Metrics
- Wrapping Up

# Individual claim reserving

An accident has occurred, the insurer has been notified, and maybe parts of the claim have already been paid, now we need to guess how much remains in this realised claim.



This is potentially a promising application for neural networks.

# Ultimate claim size

For claim  $k$ , we want to know the *ultimate claim size*

$$U_k := \sum_t P_{k,t},$$

This is the sum of all *incremental payments*  $P_{k,t}$  (also called *partial payments*) over the complete lifetime of the claim.

## *i* Assumption: payments are positive

We assume  $P_{k,t} \geq 0$  throughout. In reality some payments are *negative* — called *recoveries* (e.g. money coming back from a third party or a salvaged vehicle). They are typically so rare and small relative to normal payments that ignoring them is reasonable (though this depends on the dataset).

# Outstanding claim amount

Say the valuation date is  $v$ , then we've seen the history of the claim  $\mathcal{H}_{k,v}$  to that time, including past payments.

So the *outstanding* (future unpaid amount) for claim  $k$  is

$$O_k(v) := \sum_{t>v} P_{k,t}.$$

If the claim is still ongoing then

$$U_k = \sum_{t \leq v} P_{k,t} + O_k(v).$$

In words:

Ultimate = Cumulative payments before  $v$  + Outstanding amount from  $v$  onwards

# Total outstanding

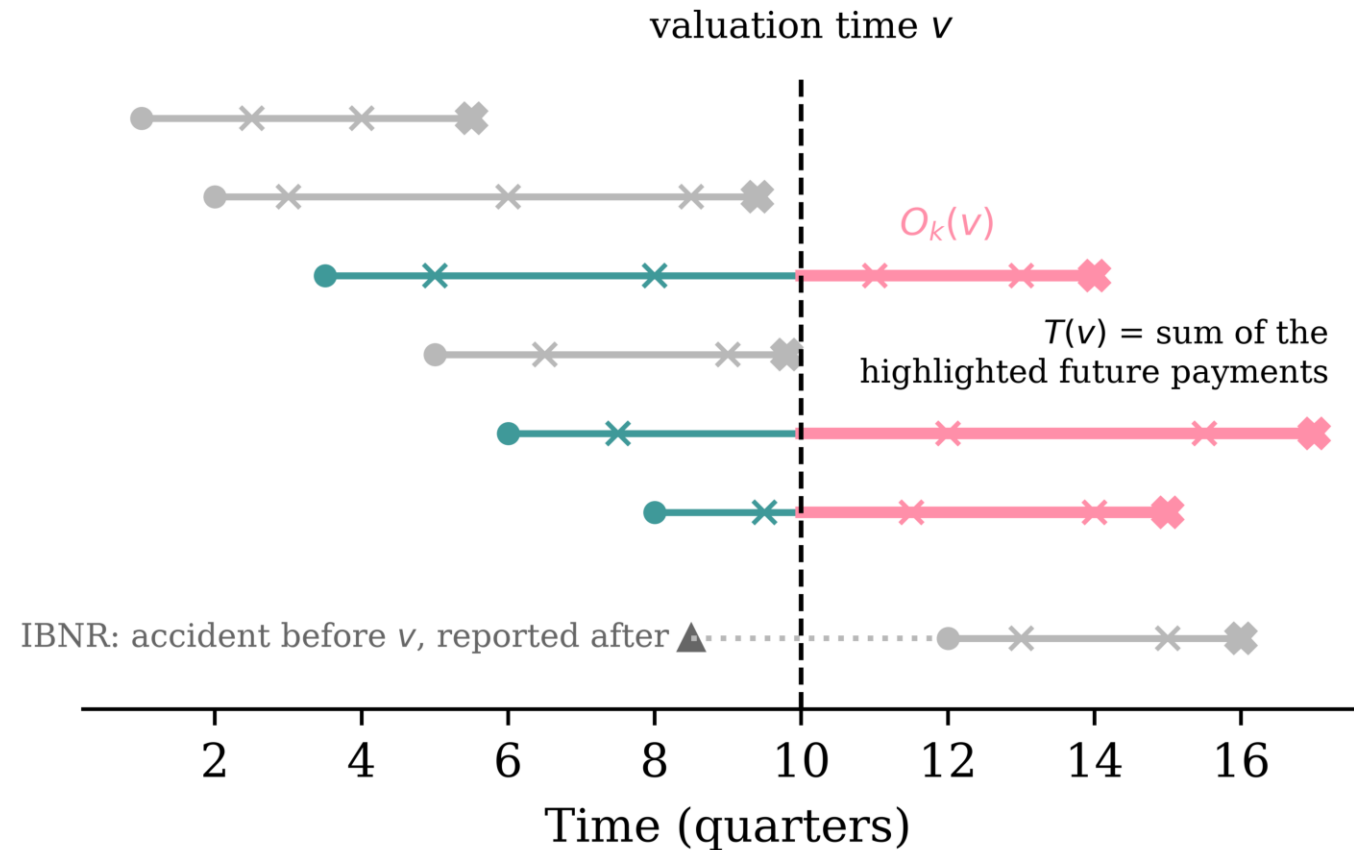
For a portfolio, we'd like to know the *total outstanding*

$$T(v) := \sum_{k \in \mathcal{O}(v)} O_k(v) ,$$

where  $\mathcal{O}(v)$  is the set of all open claims at valuation time  $v$ .

Note: this doesn't include *incurred but not reported (IBNR)* claims.

# The total outstanding, visually



Each open claim's future payments (highlighted) get summed to give  $T(v)$ .

The bottom claim has *occurred* before  $v$  but hasn't been *notified* yet. No individual model can reserve for a claim it cannot see — IBNR needs a separate adjustment.

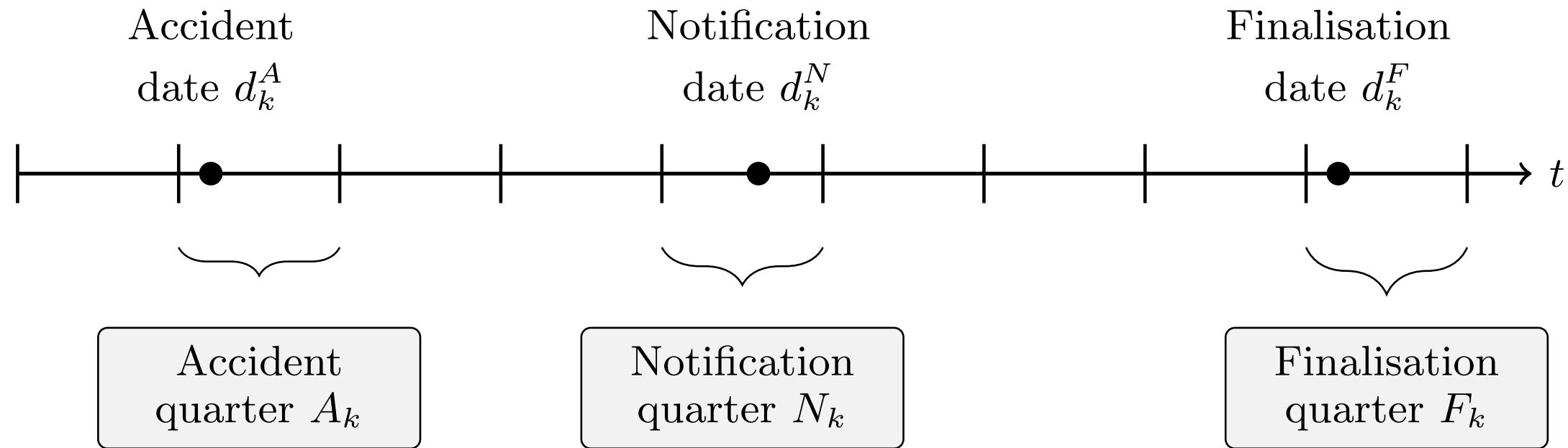
# Why is pricing different from reserving?

- *Pricing* is prospective and per-policy: no claim exists yet, and we predict the frequency/severity of hypothetical future claims from policy covariates (recall the French motor lectures).
- *Reserving* is about claims that have already happened: we condition on each claim's unfolding history — payments to date, case estimates — to predict what is still to be paid.
- Consequence for the ML setup: pricing rows are policies with static features; reserving rows are claim-quarter snapshots with *time-varying* features. That's why this lecture needs a whole preprocessing pipeline.

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# The key dates



Key dates

A claim is *open* if not yet finalised.

$$\mathcal{O}(v) := \{k : d_k^F > v\}.$$

(Though, technically, claims sometimes finalise, and then get reopened with further payments, and refinalised again.)

# Series of incremental payments

| Event        | Date       | Amount  |
|--------------|------------|---------|
| Accident     | 2021-09-30 |         |
| Notification | 2021-11-15 |         |
| Payment #1   | 2022-05-17 | \$9.00  |
| Payment #2   | 2022-08-31 | \$2.00  |
| Payment #3   | 2022-10-23 | \$2.00  |
| Payment #4   | 2022-12-20 | \$4.00  |
| Payment #5   | 2023-02-19 | \$2.00  |
| Payment #6   | 2023-05-31 | \$3.00  |
| Payment #7   | 2023-07-08 | \$18.00 |
| Payment #8   | 2023-09-10 | \$8.00  |
| Finalisation | 2023-10-28 |         |

# Case estimates

Alongside payments, the insurer holds a *case estimate*  $C_{k,v} \geq 0$ : a claims assessor's live, subjective estimate of the outstanding liability at the end of quarter  $v$ .

The assessor is aiming for

$$C_{k,v} \approx O_k(v) .$$

The crucial difference:  $C_{k,v}$  is *known at time  $v$* , whereas  $O_k(v)$  is only knowable in retrospect, once the claim finalises.

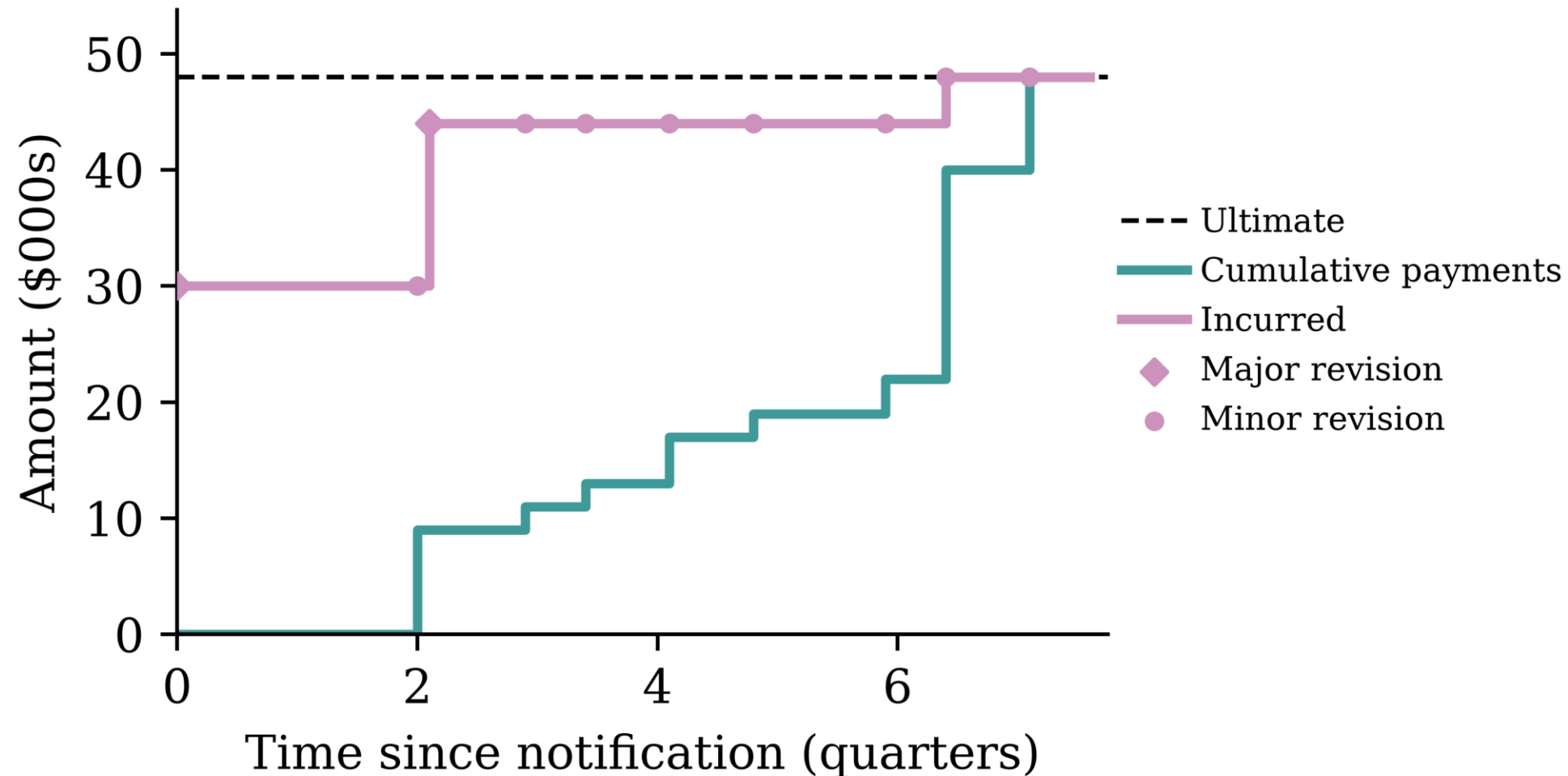
So case estimates play two roles for us:

- a *strong predictor* to feed into the model, and
- a *benchmark to beat* (if the model can't out-predict the assessors, why bother?).

# Payments and case estimates together

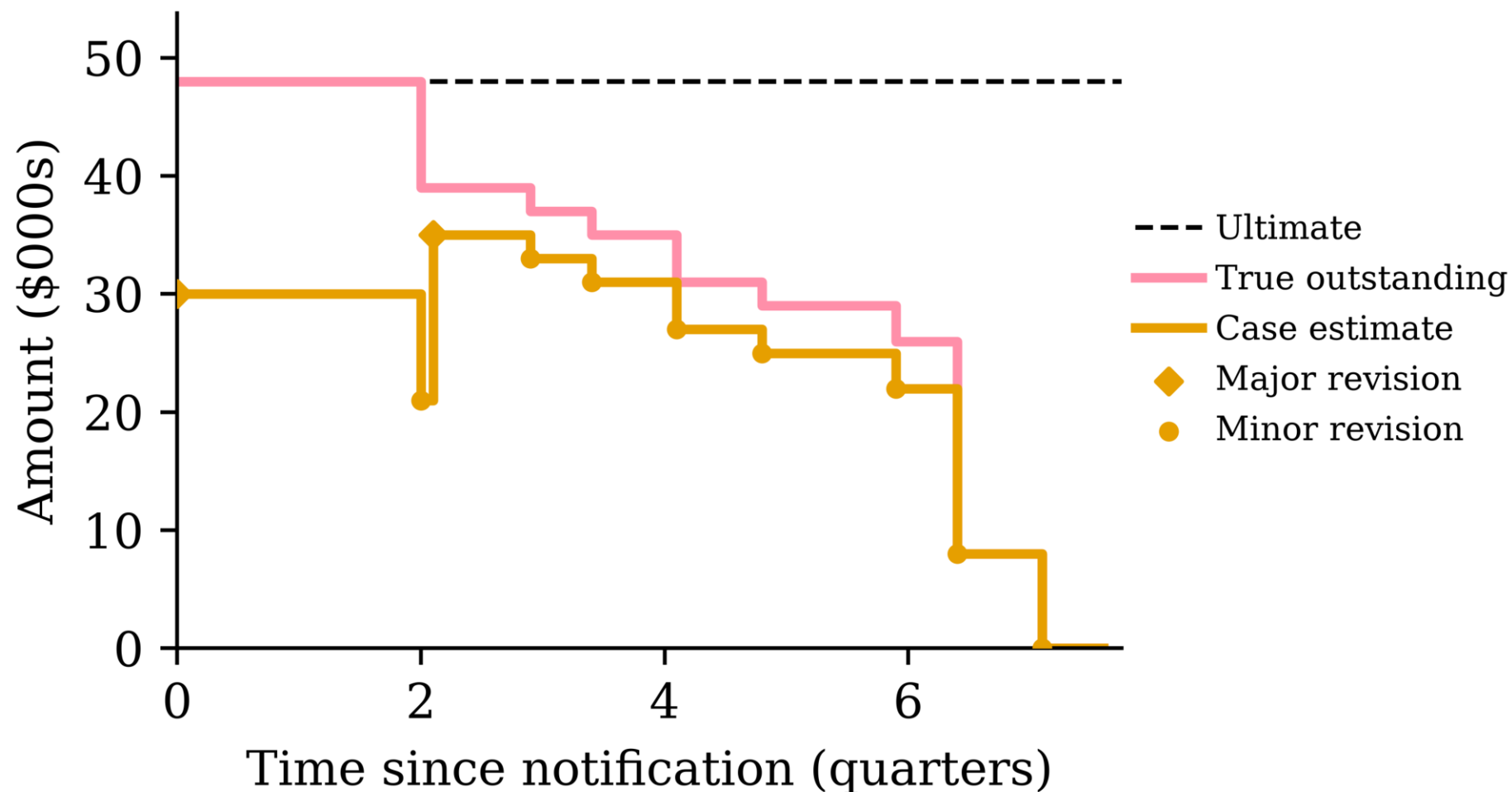
| Date       | Event                          | Payment | Case estimate |
|------------|--------------------------------|---------|---------------|
| 2021-11-15 | Notification, initial estimate |         | \$30.00       |
| 2022-05-17 | Payment #1                     | \$9.00  | \$21.00       |
| 2022-06-20 | Review: worse than hoped       |         | \$35.00       |
| 2022-08-31 | Payment #2                     | \$2.00  | \$33.00       |
| 2022-10-23 | Payment #3                     | \$2.00  | \$31.00       |
| 2022-12-20 | Payment #4                     | \$4.00  | \$27.00       |
| 2023-02-19 | Payment #5                     | \$2.00  | \$25.00       |
| 2023-05-31 | Payment #6                     | \$3.00  | \$22.00       |
| 2023-07-08 | Payment #7 (settlement agreed) | \$18.00 | \$8.00        |
| 2023-09-10 | Payment #8                     | \$8.00  | \$0.00        |
| 2023-10-28 | Finalisation                   |         | \$0.00        |

# From the policyholder's perspective



Payments step up toward the ultimate. The *incurred*, which is cumulative payments plus the case estimate of the outstanding, is the assessor's running view of the ultimate.

# From the insurer's perspective



The true outstanding  $O_k(v)$  — the *outstanding claims liability (OCL)* — and the *case estimate*  $C_{k,v}$  (the assessor's live forecast of it) both drop to zero.

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# Three levels of claim data granularity

1. *Individual transaction level*: Stores every transaction or other update for every claim. Highest granularity. Often just visible in the insured's bank statements, or in insurance company's finance records. Usually considered too granular to be useful for actuaries.
2. *Individual quarterly summaries*: This aggregates all the payments within a quarter (alternatively per month) for each claim. This is what the actuary would see.
3. *Portfolio quarterly summaries*: Aggregates all payments within a quarter (alternatively per month) across all claims. This feeds into a Chain Ladder style of reserving.

# Aggregate reserving

You have seen the classical approach to reserving already. The insurer's past payments are aggregated into a *run-off triangle*: rows are accident years, columns are development years, and each cell holds the cumulative amount paid so far.

| Accident Year | Dev 1 | Dev 2 | Dev 3 |
|---------------|-------|-------|-------|
| 2021          | 100   | 150   | 165   |
| 2022          | 120   | 180   | ?     |
| 2023          | 140   | ?     | ?     |

The lower-right of the triangle is the future: it hasn't happened yet, and the *reserve* is our estimate of it.

# Chain ladder

Chain ladder fills in the triangle by assuming each accident year develops in the same proportions as the past ones.

- Development factor from Dev 1 to Dev 2:  $f_1 = \frac{150+180}{100+120} = 1.5$
- Development factor from Dev 2 to Dev 3:  $f_2 = \frac{165}{150} = 1.1$

Projecting forward:

| Accident Year | Dev 1 | Dev 2      | Dev 3      | Reserve |
|---------------|-------|------------|------------|---------|
| 2021          | 100   | 150        | 165        | 0       |
| 2022          | 120   | 180        | <i>198</i> | 18      |
| 2023          | 140   | <i>210</i> | <i>231</i> | 91      |

The total reserve here is  $18 + 91 = 109$ .

# What chain ladder ignores

Every claim in the same accident year is treated identically. The triangle has already thrown away:

- *who* was injured (age, injury severity),
- *how* the claim is progressing (legal representation, payment pattern),
- the case estimates set by claims assessors.

If two claims have paid out the same amount so far, chain ladder implicitly reserves the same for both, even if one is a minor injury about to finalise and the other is heading to court.

# Why individual claim reserving?

With claim-level data and machine learning, we can use those discarded covariates to predict each claim's outstanding amount separately.

- The portfolio reserve is then just a sum:  $T(v) = \sum_{k \in \mathcal{O}(v)} O_k(v)$ .
- Per-claim predictions are useful beyond the total: claims triage (which claims need senior assessors?), segmenting the reserve by injury severity or legal representation, spotting claims whose case estimates look off.
- Chain ladder remains the *benchmark*: an individual model that can't beat the triangle isn't worth its complexity.

## Note

Next week's guest lecture digs into *when and why* aggregate models break down, and the trade-offs between micro- and macro-level models. Today we focus on individual-level approach and how to implement it.

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# A few major choices upfront

## 1. What are the model inputs and outputs?

Predictions are made from the perspective of some fixed time  $v$ , the *valuation quarter*.

The individual claim model has:

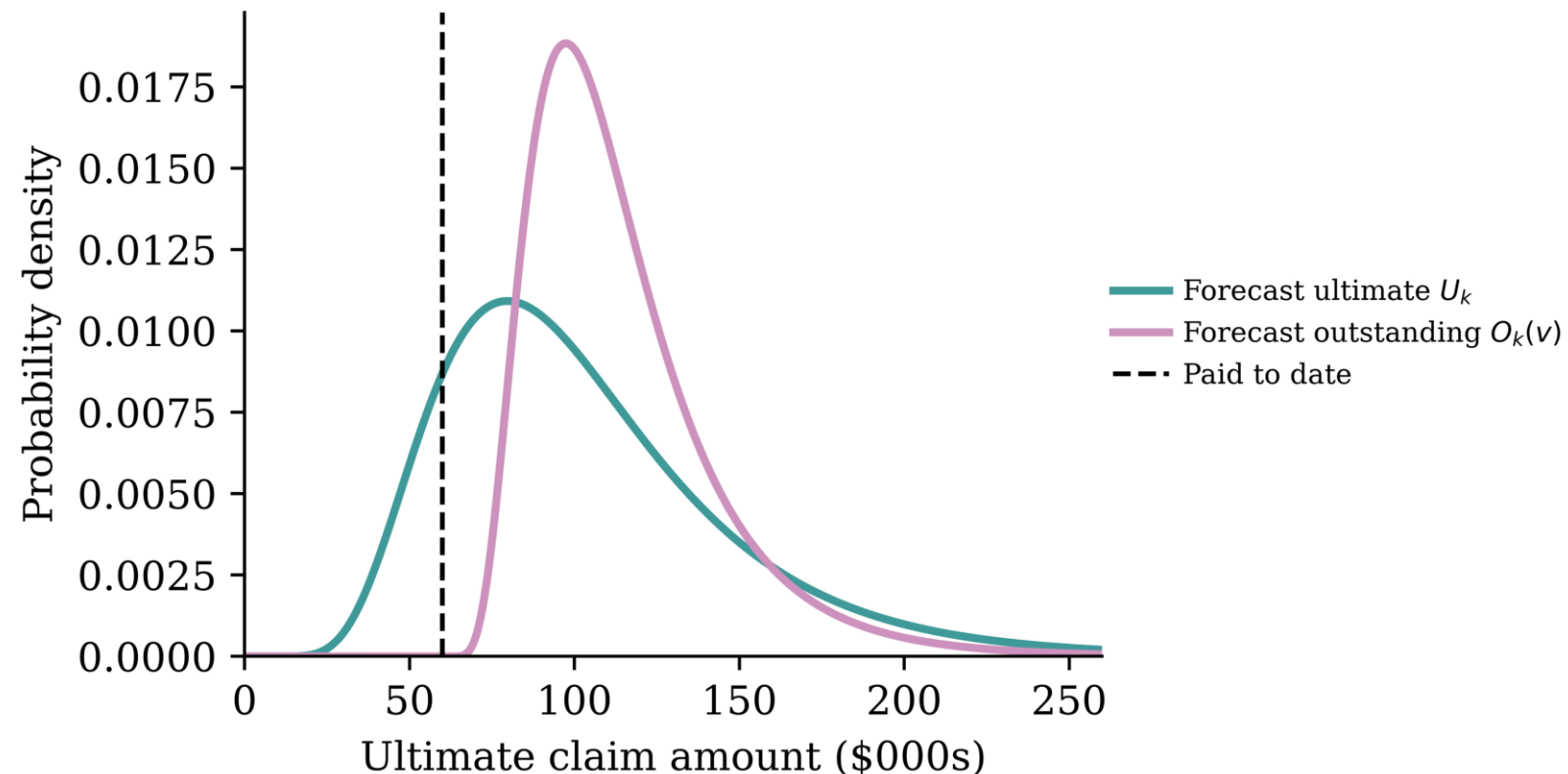
- *Input*: an  $X_{k,v}$  summarising the claim's history up to  $v$ .
- *Output*: either the ultimate claim size  $U_k$  or the outstanding value  $O_k(v)$ .

## 2. Are we performing traditional regression or distributional regression?

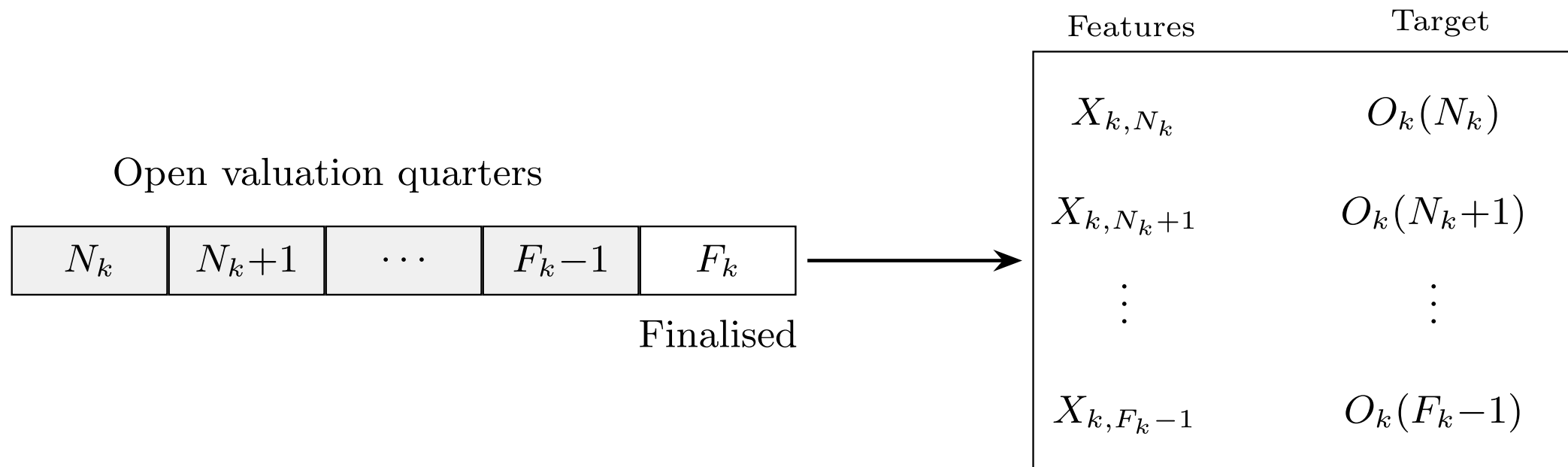
# Two potential targets: ultimate or outstanding

The output distribution can sit on either of two quantities:

- the *ultimate*  $U_k$ , or
- the *outstanding*  $O_k(v) = U_k - \sum_{t \leq v} P_{k,t}$ .



# Claim snapshots



We will turn one claim into multiple “claim snapshots”/rows of input.

One finalised claim, viewed retrospectively from *every* quarter it was open, yields one training example:

$$(X_{k,v}, O_k(v)) \text{ for } v = N_k, \dots, F_k - 1.$$

Note, claims with  $N_k = F_k$  just disappear.

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# One claim's full history

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## Static Covariates

**Type:** Motor**Postcode:** 1000**Region:** Regional**Notify Lag:** 1 quarters

# Noting the quarters

Each event date is binned into its calendar quarter ( $A_k, N_k, F_k$ ), and we also track the *development quarter* (quarters since the accident).

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| Event        | Date       | Calendar Quarter | Amount  | Dev Quarter |
|--------------|------------|------------------|---------|-------------|
| Accident     | 2021-09-30 | 2021Q3           |         | Q1          |
| Notification | 2021-11-15 | 2021Q4           |         | Q2          |
| Payment #1   | 2022-05-17 | 2022Q2           | \$9.00  | Q4          |
| Payment #2   | 2022-08-31 | 2022Q3           | \$2.00  | Q5          |
| Payment #3   | 2022-10-23 | 2022Q4           | \$2.00  | Q6          |
| Payment #4   | 2022-12-20 | 2022Q4           | \$4.00  | Q6          |
| Payment #5   | 2023-02-19 | 2023Q1           | \$2.00  | Q7          |
| Payment #6   | 2023-05-31 | 2023Q2           | \$3.00  | Q8          |
| Payment #7   | 2023-07-08 | 2023Q3           | \$18.00 | Q9          |
| Payment #8   | 2023-09-10 | 2023Q3           | \$8.00  | Q9          |
| Finalisation | 2023-10-28 | 2023Q4           |         | Q10         |

# Quarterly aggregation

All payments within a quarter collapse into one number,  $P_{k,t}$ . Case estimates within a quarter are instead summarised by their *latest* value.

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Payment Composition

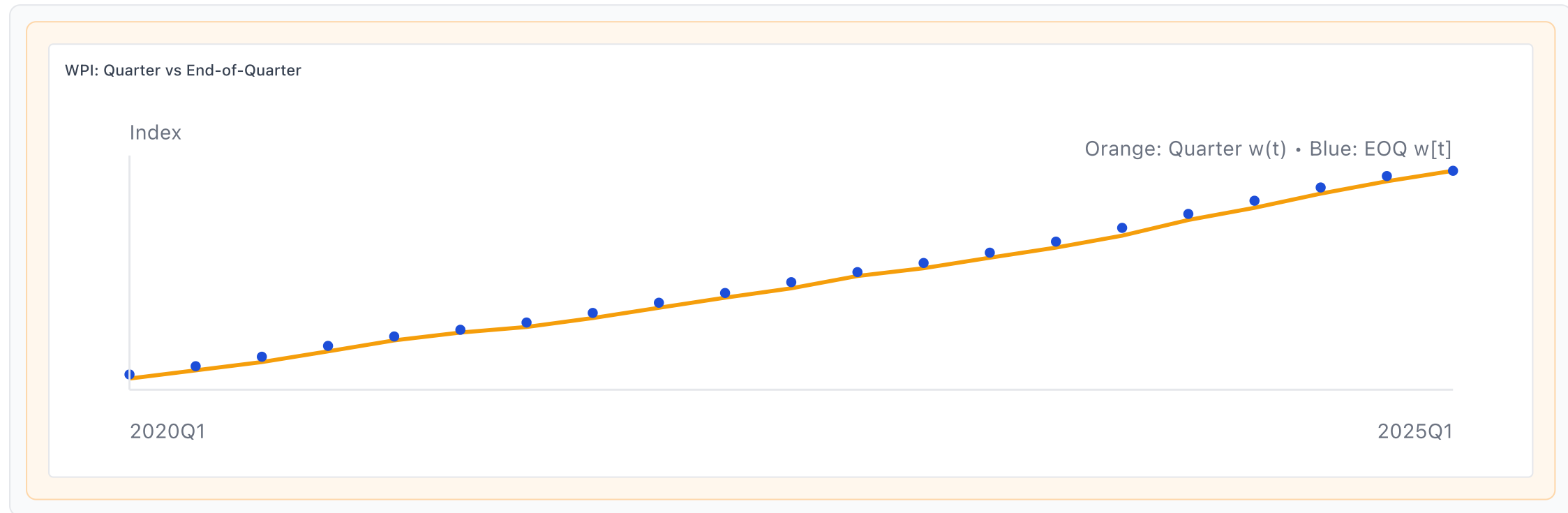
|         |        |                |
|---------|--------|----------------|
| Dev Q1  | 2021Q3 |                |
| Dev Q2  | 2021Q4 |                |
| Dev Q3  | 2022Q1 |                |
| Dev Q4  | 2022Q2 | \$9.00         |
| Dev Q5  | 2022Q3 | \$2.00         |
| Dev Q6  | 2022Q4 | \$2.00 \$4.00  |
| Dev Q7  | 2023Q1 | \$2.00         |
| Dev Q8  | 2023Q2 | \$3.00         |
| Dev Q9  | 2023Q3 | \$18.00 \$8.00 |
| Dev Q10 | 2023Q4 |                |

Quarterly Summary

| Quarter | Sum     |
|---------|---------|
| Dev Q1  | \$0.00  |
| Dev Q2  | \$0.00  |
| Dev Q3  | \$0.00  |
| Dev Q4  | \$9.00  |
| Dev Q5  | \$2.00  |
| Dev Q6  | \$6.00  |
| Dev Q7  | \$2.00  |
| Dev Q8  | \$3.00  |
| Dev Q9  | \$26.00 |
| Dev Q10 | \$0.00  |

# Adjusting for inflation

We need to adjust all monetary units to a common date to remove inflation.



# Adjusting for inflation, applied

Each of the selected claim's quarters gets an adjustment factor from the index.

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## Quarter & EOQ values and adjustment factors

| Source Quarter | w(t) source | w[t] source | Target Quarter | w[T] target | Payments: w[T]/w(t) | Case est.: w[T]/w[t] |
|----------------|-------------|-------------|----------------|-------------|---------------------|----------------------|
| 2021Q3         | 111.09      | 111.90      | 2025Q1         | 139.63      | 1.2569              | 1.2478               |
| 2021Q4         | 112.72      | 113.66      | 2025Q1         | 139.63      | 1.2387              | 1.2285               |
| 2022Q1         | 114.60      | 115.52      | 2025Q1         | 139.63      | 1.2184              | 1.2087               |
| 2022Q2         | 116.45      | 117.30      | 2025Q1         | 139.63      | 1.1991              | 1.1904               |
| 2022Q3         | 118.16      | 119.28      | 2025Q1         | 139.63      | 1.1817              | 1.1706               |
| 2022Q4         | 120.40      | 121.12      | 2025Q1         | 139.63      | 1.1597              | 1.1529               |
| 2023Q1         | 121.83      | 122.79      | 2025Q1         | 139.63      | 1.1461              | 1.1372               |
| 2023Q2         | 123.75      | 124.67      | 2025Q1         | 139.63      | 1.1283              | 1.1200               |
| 2023Q3         | 125.60      | 126.68      | 2025Q1         | 139.63      | 1.1117              | 1.1022               |
| 2023Q4         | 127.78      | 129.20      | 2025Q1         | 139.63      | 1.0927              | 1.0807               |

## Quarter-Level Adjustments

| Dev Quarter | Calendar Quarter | Nominal Sum | Adj Factor | Adjusted Sum |
|-------------|------------------|-------------|------------|--------------|
| Dev Q1      | 2021Q3           | \$0.00      | —          | \$0.00       |
| Dev Q2      | 2021Q4           | \$0.00      | —          | \$0.00       |
| Dev Q3      | 2022Q1           | \$0.00      | —          | \$0.00       |
| Dev Q4      | 2022Q2           | \$9.00      | 1.1991     | \$10.79      |
| Dev Q5      | 2022Q3           | \$2.00      | 1.1817     | \$2.36       |

# Calculate inflation-adjusted outstanding

Compute the outstanding  $O_k(v)$  retrospectively at every quarter  $v$  it was open.

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**Ultimate = Total Payments Over Claim Lifetime (adjusted to 2025Q1) = \$55.12**

| Quarter | Cumulative Paid | Outstanding Liability |
|---------|-----------------|-----------------------|
| Dev Q1  | \$0.00          | \$55.12               |
| Dev Q2  | \$0.00          | \$55.12               |
| Dev Q3  | \$0.00          | \$55.12               |
| Dev Q4  | \$10.88         | \$44.24               |
| Dev Q5  | \$13.26         | \$41.86               |
| Dev Q6  | \$20.28         | \$34.84               |
| Dev Q7  | \$22.59         | \$32.53               |
| Dev Q8  | \$26.00         | \$29.12               |
| Dev Q9  | \$55.12         | \$0.00                |
| Dev Q10 | \$55.12         | \$0.00                |

True outstanding = Ultimate – CumulativePaidToDate

# From history to features

Everything known about claim  $k$  at valuation quarter  $v$ :

$$\mathcal{H}_{k,v} := \{A_k, N_k, (Z_{k,t})_{t \leq v}, (P_{k,t})_{t \leq v}, (C_{k,t})_{t \leq v}\}$$

Some covariates  $Z_{k,t}$  are *static* (age at accident, injury severity), others are *time-varying* (legal representation can switch on mid-claim).

Problem: this history has a *different length* for every  $(k, v)$ . Tabular neural networks want a fixed-length vector.

The preprocessing pipeline is a deterministic map

$$X_{k,v} := \Phi(\mathcal{H}_{k,v}) \in \mathbb{R}^p.$$

# Two kinds of time-varying covariates

When  $\Phi$  compresses the history, the time-varying covariates split into two kinds:

1. *Only the latest value matters.* E.g. legal representation status, or the current case estimate: keep  $Z_{k,v}$ , discard the path. (Implicitly a *Markov-style* assumption: the current state is a sufficient summary.)
2. *The path matters.* E.g. the payments-per-quarter series: a claim that paid \$40 in one early lump differs from one that dribbled out \$10 four times. These get summarised as functions of the whole history — they are *path-dependent* features.

Deciding which kind each covariate is *is itself a modelling choice*: treating the case estimate as “latest value only” throws away how often the assessor has revised it.

# Summarising the payment history

How does  $\Phi$  compress a variable-length payment history into fixed columns? Summary statistics:

- development quarter  $v - A_k$ , notification delay  $N_k - A_k$ ,
- count / sum / mean / std / min / max of past payments,
- the latest case estimate,
- current values of the time-varying covariates,
- the static covariates unchanged.

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Choose valuation quarter:

Dev Q2

Dev Q11

Valuation: Dev Q6 2022Q3

Near-Static Covariates (latest up to cutoff)

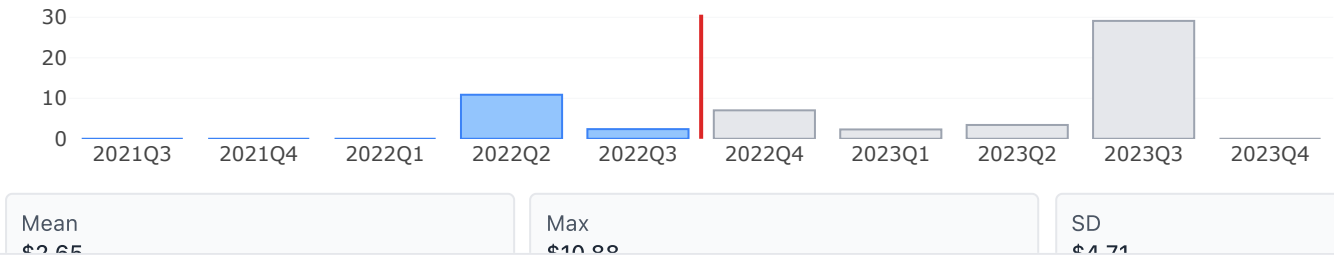
| Dev Q | Calendar Q | Postcode | Legal Rep |
|-------|------------|----------|-----------|
| Q2    | 2021Q3     | 1000     | NA        |
| Q3    | 2021Q4     | 1000     | NA        |
| Q4    | 2022Q1     | 1000     | Yes       |
| Q5    | 2022Q2     | 1000     | Yes       |
| Q6    | 2022Q3     | 1000     | Yes       |
| Q7    | 2022Q4     | 1000     | Yes       |

Most recent Postcode  
1000

Claimant has legal representation  
Yes

Time-Series Covariates (summary up to cutoff)

Incremental payments



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Choose valuation quarter:

Dev  
Q2

Valuation: Dev Q6 2022Q3

Dev  
Q11

Training Row (as-of Dev Q6)

| Feature             | Type          | Rule                                         | Value          |
|---------------------|---------------|----------------------------------------------|----------------|
| postcode_latest     | Near-static   | Latest non-missing $\leq$ cutoff             | 1000           |
| legal_rep_latest    | Near-static   | Latest non-missing $\leq$ cutoff             | Yes            |
| inc_paid_mean_q1..k | Time-series   | Mean of increments $\leq$ cutoff             | \$2.65         |
| inc_paid_max_q1..k  | Time-series   | Max of increments $\leq$ cutoff              | \$10.88        |
| inc_paid_sd_q1..k   | Time-series   | Std dev of increments $\leq$ cutoff          | \$4.71         |
| cum_paid_last_qk    | Time-series   | Last observed $\leq$ cutoff                  | \$13.26        |
| case_estimate       | Time-series   | Case estimate (latest $\leq$ cutoff)         | \$44.40        |
| <b>outstanding</b>  | <b>Target</b> | Actual outstanding = ultimate – paid-to-date | <b>\$41.86</b> |

# Alternative: let an RNN read the history

The claim history is naturally a *sequence* — one vector per development quarter:

$$(P_{k,t}, C_{k,t}, Z_{k,t}) \quad \text{for } t = N_k, \dots, v.$$

So instead of hand-crafting summary statistics, we could:

- feed the quarter-by-quarter vectors into a recurrent layer (e.g. GRU/LSTM),
- take the final hidden state as a *learned* summary of the history,
- concatenate the static covariates, and predict  $O_k(v)$  with a dense head.

# Preparing features for a neural network

Finally, add standard NN preprocessing ( $\log(1 + x)$  transforming monetary values), encoding categoricals, normalising continuous features.

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Choose valuation quarter:

Dev Q2

Valuation: Dev Q6 2022Q3

Dev Q11

Training Row with Log-Transformed Dollar Features

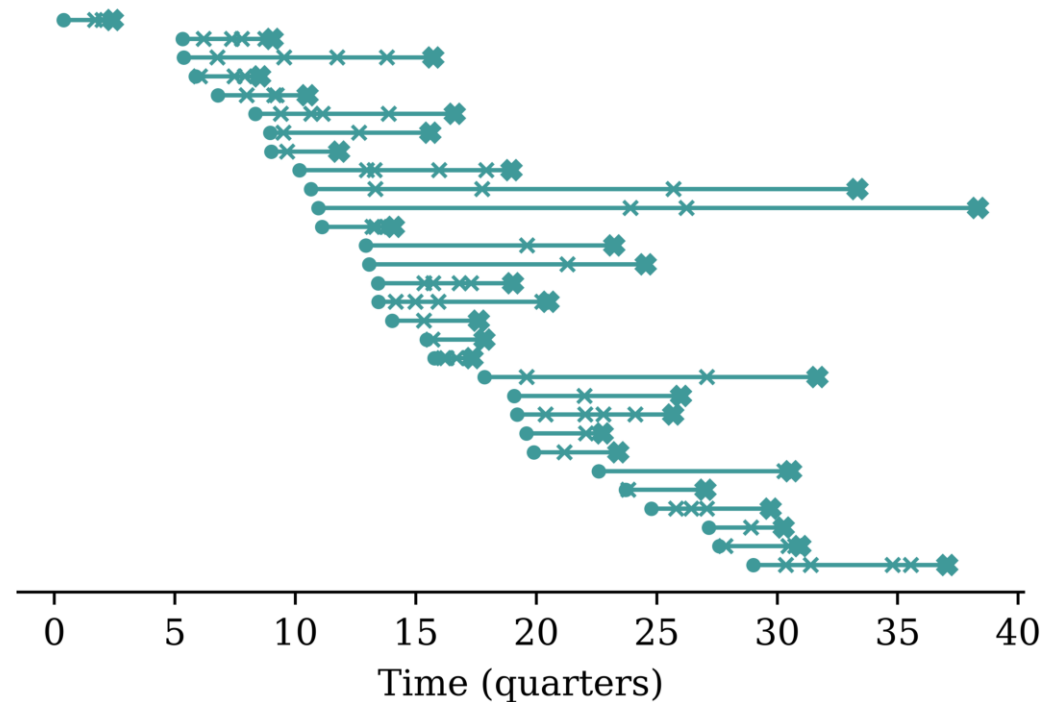
| Feature             | Original Value                          | $\log(1 + x)$ |
|---------------------|-----------------------------------------|---------------|
| postcode_latest     | (categorical - will be one-hot encoded) |               |
| legal_rep_latest    | (categorical - will be one-hot encoded) |               |
| inc_paid_mean_q1..k | \$2.65                                  | 1.2952        |
| inc_paid_max_q1..k  | \$10.88                                 | 2.4747        |
| inc_paid_sd_q1..k   | \$4.22                                  | 1.6516        |
| cum_paid_last_qk    | \$13.26                                 | 2.6574        |
| case_estimate       | \$37.45                                 | 3.6492        |
| <b>outstanding</b>  | <b>\$41.86</b>                          |               |

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# A portfolio of claims over time

Now zoom out from one claim to the whole portfolio.

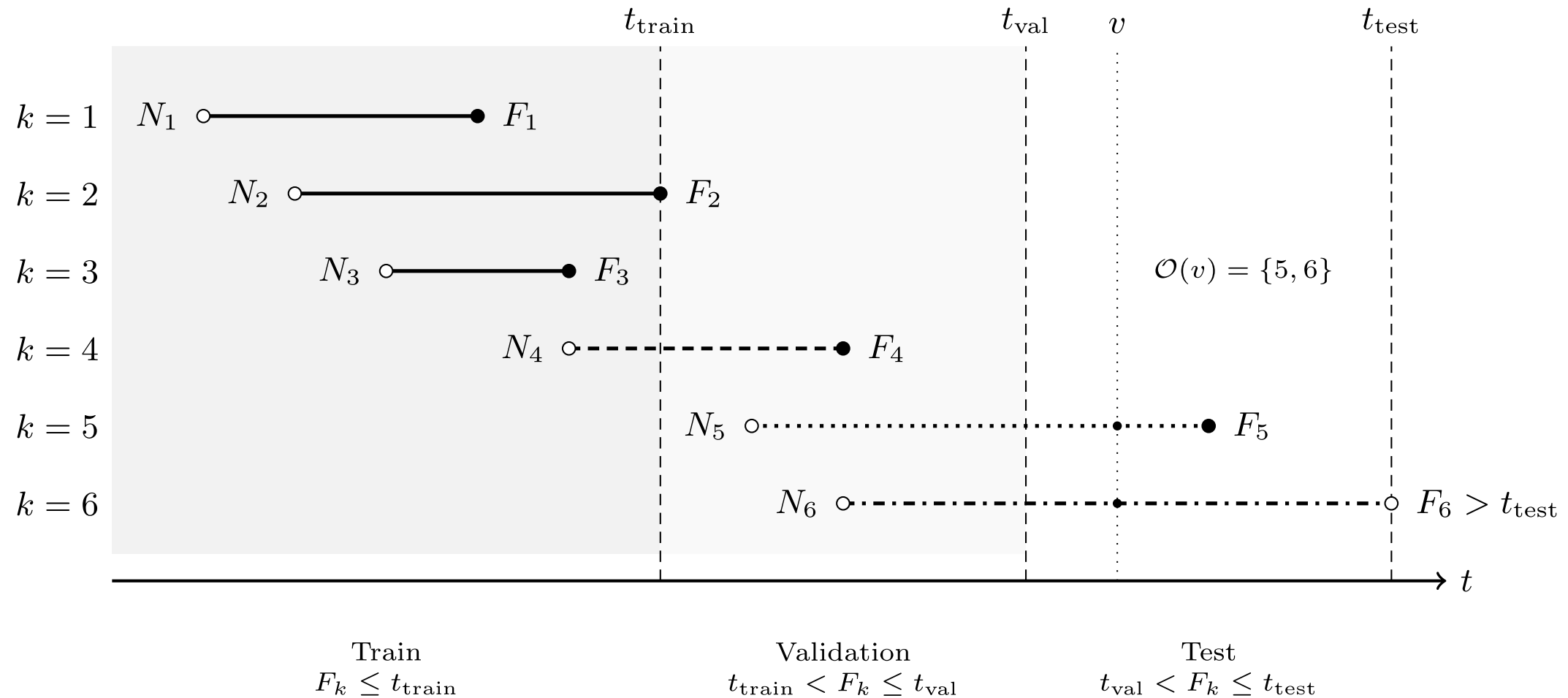


Every claim needs to be allocated to train, validation, or test — but claims *overlap in time*.

Remember, claim  $k$  generates  $F_k - N_k$  snapshot rows, and consecutive snapshots of the same claim are *near-duplicates* (the history only grows by one quarter). So: *all snapshots from one claim go to the same set*.

# The simplest approach

Order claims by finalisation quarter, then cut into *train*, *validation*, and *test* eras:

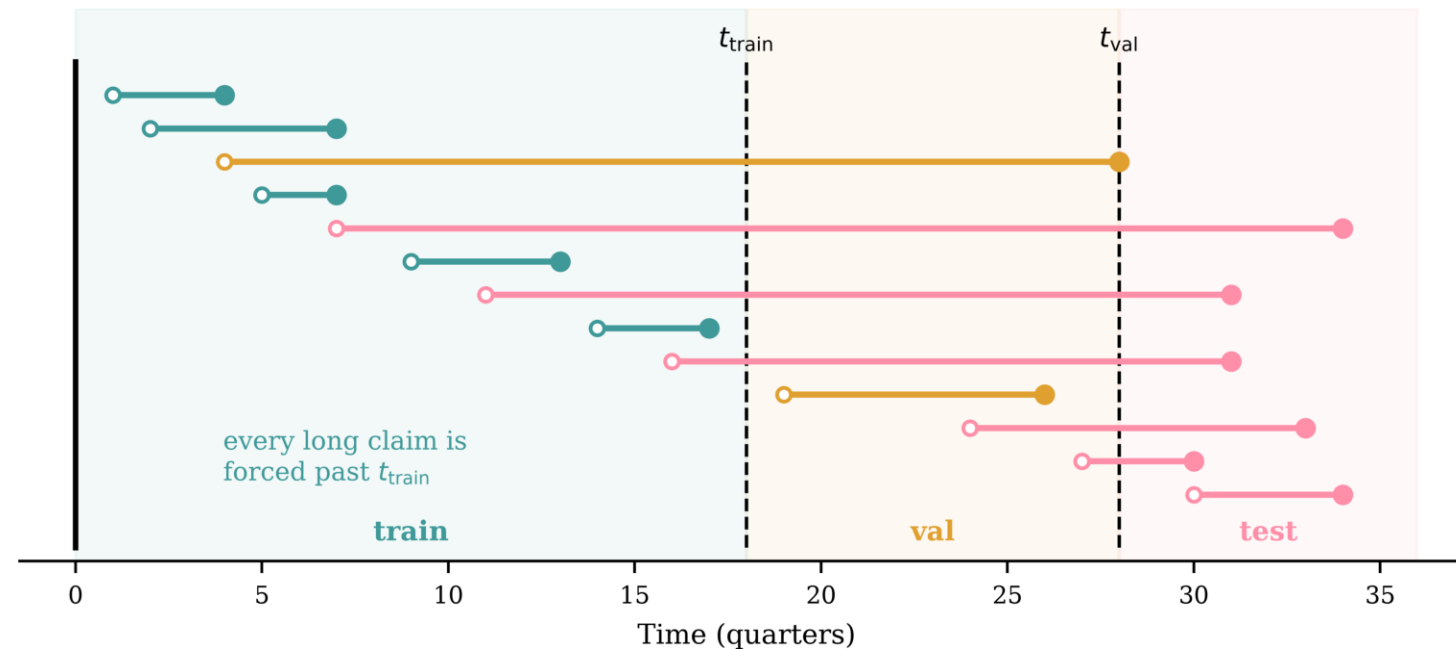


Those still open at  $t_{\text{test}}$  are the *unfinalised* set.

# Old claims skew a naive split

Suppose we keep every claim *notified* after the portfolio's inception — the obvious thing to do for a *brand-new* insurer. A hard wall then sits at the start: no claim reaches back across it.

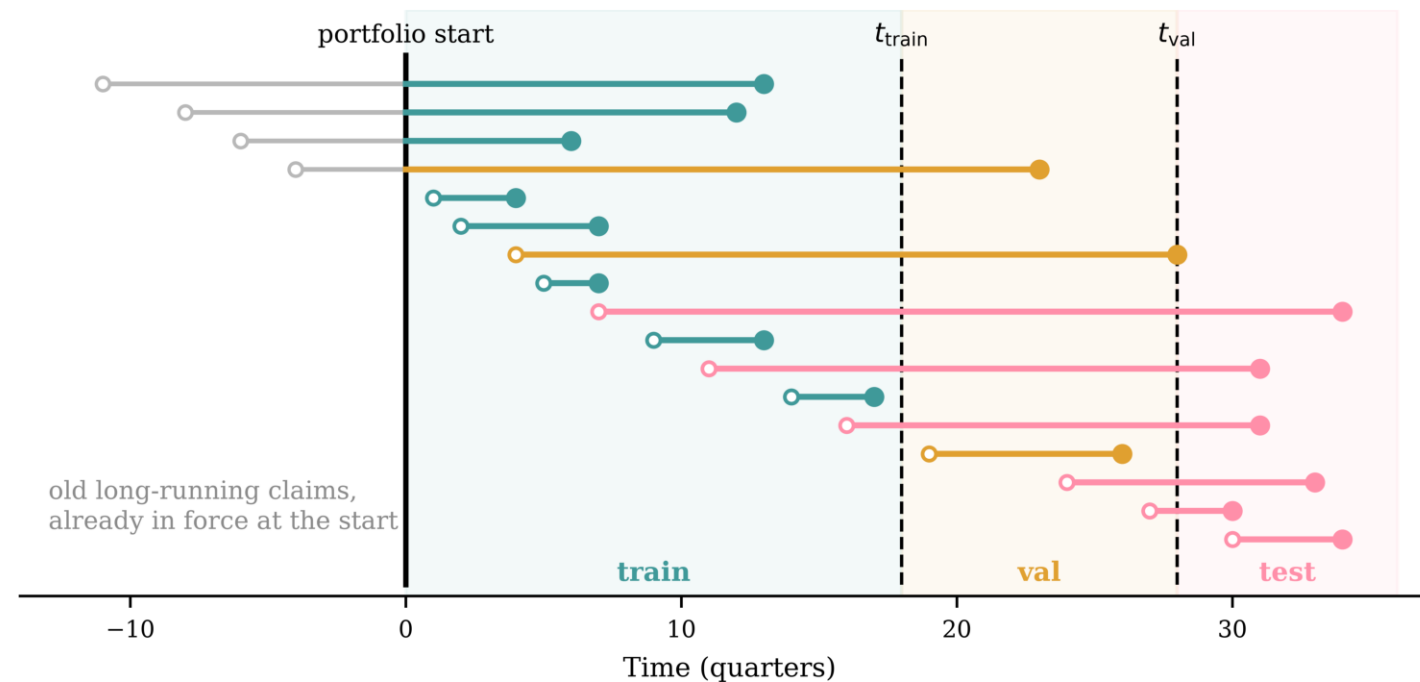
A claim can only finalise in the *train* era if it both *arrived* and *closed* early, which caps its duration at  $t_{\text{train}}$ . Every genuinely long-running claim is forced past the cut, into *validation* or *test*.



The *train* set never sees a long claim, yet the model is judged on them — a mismatch

# Fix: arrive at a steady-state portfolio

A real reserving team never starts at inception; they inherit a portfolio already *in force*, full of claims opened years earlier. We mimic that: simulate a long *burn-in* of history first, then keep every claim *finalised* after a chosen start date (not every claim *notified* after it).

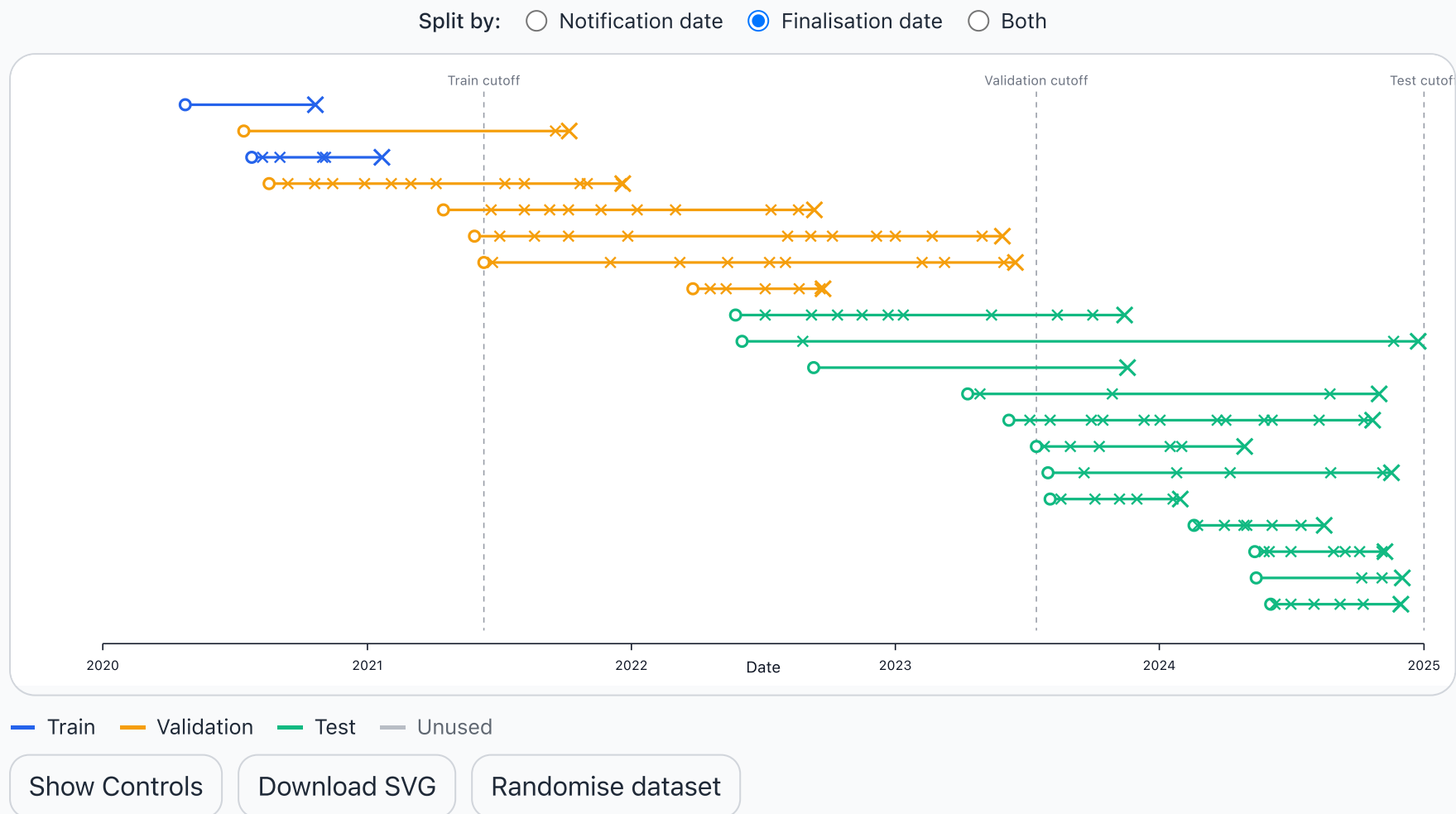


Old long-running claims now finalise inside the *train* era too, so every split carries the full spread of durations. We simulate many quarters and place the observation window deep in the steady state.

# Lecture Outline

- Introduction
- Insurance Claim Payments and Case Estimates
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- Individual Claim Modelling Choices
- Preprocessing One Claim
- Splitting Claims by Finalisation Quarter
- **Splitting Claims by Notification Quarter (Advanced)**
- Case Study: Synthetic Claims
- Fitting the Models
- Individual-level Metrics
- Portfolio-level Metrics
- Wrapping Up

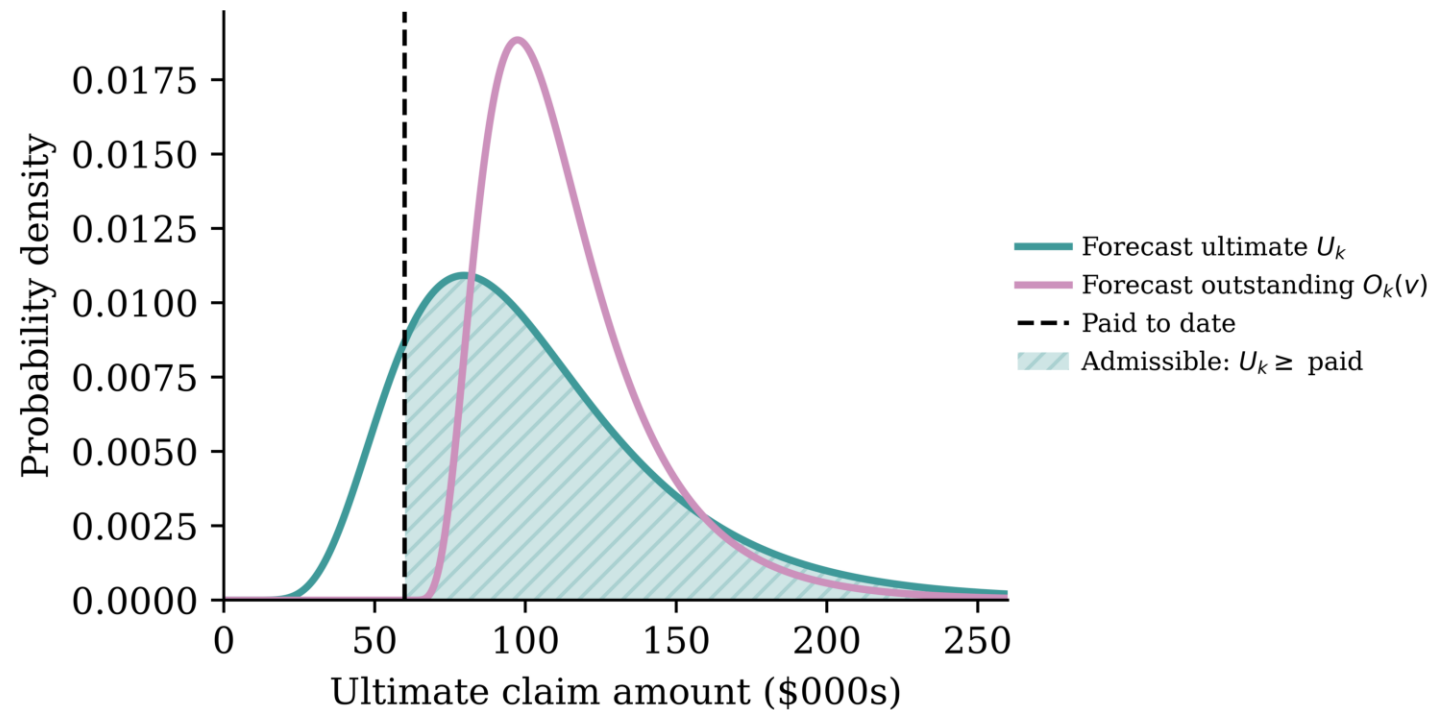
# This is more complex



# Never-finalising claims are censored

If the claim  $k$  is still open at the end of our dataset, then we have not observed the ultimate.

We only know a lower bound,  $U_k \geq \sum_{t \leq v} P_{k,t}$ , i.e. it is a *right-censored* observation (the survival-analysis kind).



The ultimate model must put *zero* mass left of the dashed line. The outstanding model gets this free: it lives on  $[0, \infty)$  from the start.

# Training directly on the censored ultimate

Keep the ultimate  $U_k$  as the target and borrow the *censored likelihood* from survival analysis.

- $\delta_k \in \{0, 1\}$ : claim  $k$  is *finalised* ( $\delta_k = 1$ ) or still *open* ( $\delta_k = 0$ ).
- $P_k(v) := \sum_{t \leq v} P_{k,t}$ : the *cumulative paid to date* for claim  $k$  at the training cut  $v$ .
- Distributional regression:  $U_k \mid X_k \sim \mathcal{D}(\vartheta_k)$  with  $\vartheta_k = \vartheta(X_k; \theta)$ , giving density  $f_k$ , cdf  $F_k$ , and survival  $S_k = 1 - F_k$ .

What we actually observe:

- *Finalised* ( $\delta_k = 1$ ): the exact ultimate  $U_k = u_k$ .
- *Open* ( $\delta_k = 0$ ): only the bound  $U_k \geq P_k(v)$  (right-censored at  $P_k(v)$ ).

# The censored log-likelihood

A finalised claim contributes its *density* at  $u_k$ ; an open claim contributes its *survival* at  $P_k(v)$  (all we know is  $U_k \geq P_k(v)$ ):

$$L(\theta) = \prod_{k=1}^n [f_k(u_k)]^{\delta_k} [S_k(P_k(v))]^{1-\delta_k}.$$

The loss is the negative log-likelihood:

$$-\ell(\theta) = -\sum_{k=1}^n \left[ \delta_k \log f_k(u_k) + (1 - \delta_k) \log (1 - F_k(P_k(v))) \right].$$

Open claims still inform  $\theta$  through the survival term  $S_k(P_k(v))$ : the constraint  $U_k \geq P_k(v)$  is genuinely informative when the target is the ultimate.

# But if we're predicting the outstanding...

We'll predict the *outstanding*  $O_k(v)$  rather than the ultimate  $U_k$  directly.

- For a censored claim, the bound  $U_k \geq P_k(v)$  becomes just  $O_k(v) \geq 0$ .
- Any sensible model of the outstanding only predicts *positive* values.
- So the censoring constraint is satisfied *by construction* — censored claims carry no usable information, and we can simply drop them.

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# Imports for the case study

The R side simulates the data and runs the classical benchmark:

```
1 library(SPLICE)
2 library(ChainLadder)
```

The Python side does the machine learning:

```
1 import numpy as np
2 import pandas as pd
3 from sklearn.compose import ColumnTransformer
4 from sklearn.linear_model import Ridge
5 from sklearn.metrics import root_mean_squared_error
6 from sklearn.pipeline import make_pipeline
7 from sklearn.preprocessing import FunctionTransformer, MinMaxScaler, OneHotEncoder
8
9 from keras.models import Sequential
10 from keras.layers import Dense
11 from keras.callbacks import EarlyStopping
```

# Generating synthetic claims with SPLICE

```

1  sim ← generate_data(
2    n_claims_per_period = 100,    # ~10,000 claims over the full horizon
3    n_periods           = 100,    # long burn-in; we observe the last ~40 quarters
4    complexity          = 2,      # development depends on claim size; 2% p.a. inflation
5    random_seed         = 20260712,
6    covariates_obj       = SynthETIC::test_covariates_obj
7  )

```

[1] "Generating claims, payments, incurred data with complexity = 2 (scenario: claim development depends on claim size) ,,,,"

Three main tables: one row per *claim*, per *payment*, and per *case estimate transaction*.

# All claims

```
1 claims ← sim$claim_dataset
```

|    | claim_no<br><int> | occurrence_period<br><dbl> |
|----|-------------------|----------------------------|
| 1  | 1                 | 1                          |
| 2  | 2                 | 1                          |
| 3  | 3                 | 1                          |
| 4  | 4                 | 1                          |
| 5  | 5                 | 1                          |
| 6  | 6                 | 1                          |
| 7  | 7                 | 1                          |
| 8  | 8                 | 1                          |
| 9  | 9                 | 1                          |
| 10 | 10                | 1                          |

1-10 of 9,974 rows | 1-3 of 8 columns

Previous

1

2

3

4

5

6

...

20

Next

# All static covariates

```
1 covariates ← sim$covariates_data$data
```

|    | Legal Representation<br><fct> |  |
|----|-------------------------------|--|
| 1  | Y                             |  |
| 2  | Y                             |  |
| 3  | Y                             |  |
| 4  | Y                             |  |
| 5  | Y                             |  |
| 6  | Y                             |  |
| 7  | Y                             |  |
| 8  | Y                             |  |
| 9  | Y                             |  |
| 10 | Y                             |  |

1-10 of 9,974 rows | 1-2 of 4 columns

Previous 1 2 3 4 5 6 ... 20 Next

# All payments

```
1 payments ← sim$payment_dataset
```

|    | claim_no<br><int> | pmt_no<br><dbl> | occurrence_period<br><dbl> |
|----|-------------------|-----------------|----------------------------|
| 1  | 1                 | 1               | 1                          |
| 2  | 1                 | 2               | 1                          |
| 3  | 1                 | 3               | 1                          |
| 4  | 1                 | 4               | 1                          |
| 5  | 2                 | 1               | 1                          |
| 6  | 2                 | 2               | 1                          |
| 7  | 2                 | 3               | 1                          |
| 8  | 2                 | 4               | 1                          |
| 9  | 2                 | 5               | 1                          |
| 10 | 3                 | 1               | 1                          |

1-10 of 49,611 rows | 1-4 of 13 columns

Previous 1 2 3 4 5 6 ... 20 Next

# All estimates

```
1 estimates ← sim$incurred_dataset
```

|    | claim_no<br><int> | claim_size<br><dbl> | txn_time<br><dbl> |
|----|-------------------|---------------------|-------------------|
| 1  | 1                 | 23985.2153          | 2.5533059         |
| 2  | 1                 | 23985.2153          | 3.4335363         |
| 3  | 1                 | 23985.2153          | 5.7702533         |
| 4  | 1                 | 23985.2153          | 6.2715888         |
| 5  | 1                 | 23985.2153          | 7.2050180         |
| 6  | 1                 | 23985.2153          | 7.5899149         |
| 7  | 1                 | 23985.2153          | 7.7703589         |
| 8  | 1                 | 23985.2153          | 8.1934959         |
| 9  | 1                 | 23985.2153          | 8.8744302         |
| 10 | 2                 | 67757.9232          | 0.3823298         |

1-10 of 84,839 rows | 1-4 of 10 columns

Previous

1

2

3

4

5

6

...

20

Next

# Minor tweaks

Add the static covariates to the claims dataframe:

```
1 claims <- cbind(claims, covariates)
```

SPLICE names the case-estimate column `OCL`; rename it to `estimate`:

```
1 names(estimates)[names(estimates) == "OCL"] <- "estimate"
```

Drop the `incurred` column, since it is just `cumpaid` plus `estimate`:

```
1 estimates$incurred <- NULL
```

SPLICE's estimates are nominal, so (for simplicity) we manually remove inflation:

```
1 estimates$estimate <- estimates$estimate * 1.02^(-ceiling(estimates$txn_time) / 4)
```

# One synthetic claim's payments

One claim's payments (note the paired nominal/deflated columns):

```
1 payments[payments["claim_no"] == 1][ ... ]
```

|          | pmt_no | payment_time | payment_period | payment_size | payment_inflated |
|----------|--------|--------------|----------------|--------------|------------------|
| <b>0</b> | 1.0    | 3.433536     | 4.0            | 3443.887887  | 3502.928359      |
| <b>1</b> | 2.0    | 7.205018     | 8.0            | 3634.989595  | 3766.988290      |
| <b>2</b> | 3.0    | 7.770359     | 8.0            | 15120.618241 | 15713.616650     |
| <b>3</b> | 4.0    | 8.874430     | 9.0            | 1785.719555  | 1865.922752      |

# One synthetic claim's case estimates

Its case estimate transactions:

```
1 estimates[estimates["claim_no"] = 1][ ... ]
```

|            | txn_time | txn_type | cumpaid      | estimate     |
|------------|----------|----------|--------------|--------------|
| <b>0</b>   | 2.553306 | Ma       | 0.000000     | 28070.664592 |
| <b>1</b>   | 3.433536 | PMi      | 3502.928359  | 25734.342075 |
| <b>2</b>   | 5.770253 | Mi       | 3502.928359  | 26269.964789 |
| <b>...</b> | ...      | ...      | ...          | ...          |
| <b>6</b>   | 7.770359 | P        | 22983.533298 | 2037.350270  |
| <b>7</b>   | 8.193496 | Mi       | 22983.533298 | 1857.027805  |
| <b>8</b>   | 8.874430 | PMi      | 24849.456051 | 0.000000     |

9 rows × 4 columns

# From transactions to snapshot rows

Bin each claim's key dates into quarters, and sort the claims into groups:

```

1  # We "arrive" at the portfolio at T_START and value it at T_END: keep claims
2  # that finalise inside this steady-state window, not those that merely start in it.
3  T_START, T_END = 60, 100
4
5  claims["notif_qtr"] = np.ceil(claims["occurrence_time"] + claims["notidel"]).astype(int)
6  claims["final_qtr"] = np.ceil(claims["occurrence_time"] + claims["notidel"] + claims["set
7
8  notif, final = claims["notif_qtr"], claims["final_qtr"]
9  is_ibnr = notif > T_END
10 is_unfinalised = (notif ≤ T_END) & (final > T_END)
11 is_flash = (final ≤ T_END) & (final == notif)
12 is_finalised = (final > T_START) & (final ≤ T_END) & (final > notif)
13
14 # The claims we model: those finalised inside the window (which carry a known
15 # target) plus those still open at T_END (which are what we must reserve for).
16 observed = claims[is_finalised | is_unfinalised]

```

9,974 claims: 3,935 finalised & usable, 971 unfinalised, 214 IBNR, 162 settled within a quarter

# Building the snapshot grid

One row per claim per open quarter, with the quarterly payments  $P_{k,t}$  merged in (missing quarters become zero-payment rows).

Before — one row per *claim*:

|             | claim_no | notif_qtr | final |
|-------------|----------|-----------|-------|
| <b>2823</b> | 2824     | 31        | 61    |
| <b>2919</b> | 2920     | 31        | 65    |
| <b>2987</b> | 2988     | 32        | 63    |

After — one row per claim per open *quarter*:

|          | claim_no | quarter | Incremental |
|----------|----------|---------|-------------|
| <b>0</b> | 2824     | 31      | 0.00        |
| <b>1</b> | 2824     | 32      | 0.00        |
| <b>2</b> | 2824     | 33      | 0.00        |
| <b>3</b> | 2824     | 34      | 0.00        |
| <b>4</b> | 2824     | 35      | 24071.86    |
| <b>5</b> | 2824     | 36      | 0.00        |

# History summaries, case estimates, and the target

Onto that grid we add the path-dependent features (expanding summaries of the payment history), the current-state features (latest case estimate, forward-filled between revisions).

Before — one claim's rows:

|          | quarter | Incremental |
|----------|---------|-------------|
| <b>0</b> | 31      | 0.00        |
| <b>1</b> | 32      | 0.00        |
| <b>2</b> | 33      | 0.00        |
| <b>3</b> | 34      | 0.00        |
| <b>4</b> | 35      | 24071.86    |

After — summaries, case estimate, and the label  $O_k(v)$ :

|          | quarter | sum_Incremental | Estimate  | Outstanding |
|----------|---------|-----------------|-----------|-------------|
| <b>0</b> | 31      | 0.00            | 106362.62 | 1573886.40  |
| <b>1</b> | 32      | 0.00            | 106362.62 | 1573886.40  |
| <b>2</b> | 33      | 0.00            | 106362.62 | 1573886.40  |
| <b>3</b> | 34      | 0.00            | 106362.62 | 1573886.40  |
| <b>4</b> | 35      | 24071.86        | 80251.97  | 1549814.53  |

# The dataset after preprocessing

```
1 grid[["claim_no", "quarter", "development_period", "Incremental",
2       "sum_Incremental", "Estimate", "Outstanding"]].head(8)
```

|     | claim_no | quarter | development_period | Incremental | sum_Incremental |     |
|-----|----------|---------|--------------------|-------------|-----------------|-----|
| 0   | 2824     | 31      | 3.0                | 0.0         | 0.000000        | 106 |
| 1   | 2824     | 32      | 4.0                | 0.0         | 0.000000        | 106 |
| 2   | 2824     | 33      | 5.0                | 0.0         | 0.000000        | 106 |
| ... | ...      | ...     | ...                | ...         | ...             | ... |
| 5   | 2824     | 36      | 8.0                | 0.0         | 24071.863793    | 802 |
| 6   | 2824     | 37      | 9.0                | 0.0         | 24071.863793    | 802 |
| 7   | 2824     | 38      | 10.0               | 0.0         | 24071.863793    | 802 |

8 rows × 7 columns

# Splitting by finalisation quarter

```

1 T_TRAIN, T_VAL = 88, 94 # grid already starts at T_START, so train is (60, 88]
2 train = grid[grid["final_qtr"] ≤ T_TRAIN]
3 val = grid[(grid["final_qtr"] > T_TRAIN) & (grid["final_qtr"] ≤ T_VAL)]
4 test = grid[(grid["final_qtr"] > T_VAL) & (grid["final_qtr"] ≤ T_END)]
5
6 # Still open at the valuation date: no known target, so these are never trained
7 # or scored on – they are the claims we must reserve for. Keep each one's
8 # valuation-date row, the single snapshot we predict from.
9 unfinalised = grid[(grid["final_qtr"] > T_END) & (grid["quarter"] = T_END)]

```

|       |             |              |                                  |
|-------|-------------|--------------|----------------------------------|
| train | 28,992 rows | 2,790 claims | (finalisation quarters 61..88)   |
| val   | 6,273 rows  | 598 claims   | (finalisation quarters 89..94)   |
| test  | 5,593 rows  | 547 claims   | (finalisation quarters 95..100)  |
| unfin | 971 rows    | 971 claims   | (finalisation quarters 101..137) |

Note rows  $\gg$  claims: each claim contributes one row per quarter it was open — and every one of a claim's rows lands in the same set. The unfinalised set is the exception: one row per claim, at the valuation date.

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# Preprocessing setup

Two versions: *without* and *with* the case estimate as a feature.

```
1 dollar_cols = ["Incremental", "sum_Incremental", "mean_Incremental",
2               "std_Incremental", "min_Incremental", "max_Incremental"]
3 numeric_cols = ["development_period", "notidel", "count_Incremental"]
4 cat_cols = ["Legal Representation", "Injury Severity", "Age of Claimant"]
5
6 def make_preprocessor(with_estimate):
7     dollars = dollar_cols + (["Estimate"] if with_estimate else [])
8     return ColumnTransformer([
9         ("dollars", make_pipeline(FunctionTransformer(np.log1p), MinMaxScaler()), dollars
10        ("numeric", MinMaxScaler(), numeric_cols),
11        ("categorical", OneHotEncoder(drop="first", sparse_output=False), cat_cols),
12    ])
```

# Setup $X$ & $y$ variables

We model  $\log O_k(v)$  rather than  $O_k(v)$  because the outstanding is *heavy-tailed*.

```
1 y_train = np.log(train["Outstanding"])
2 y_val = np.log(val["Outstanding"])
3 y_test = np.log(test["Outstanding"])
4
5 ct_plain = make_preprocessor(with_estimate=False)
6 X_train_plain = ct_plain.fit_transform(train)
7 X_val_plain = ct_plain.transform(val)
8 X_test_plain = ct_plain.transform(test)
9 X_unfinalised_plain = ct_plain.transform(unfinalised)
10
11 ct_est = make_preprocessor(with_estimate=True)
12 X_train_est = ct_est.fit_transform(train)
13 X_val_est = ct_est.transform(val)
14 X_test_est = ct_est.transform(test)
15 X_unfinalised_est = ct_est.transform(unfinalised)
```

# Fitting individual claim models

```
1 glm_plain = Ridge(alpha=1e-4).fit(X_train_plain, y_train)
2 glm_est = Ridge(alpha=1e-4).fit(X_train_est, y_train)
```

```
1 nn = Sequential([
2     Dense(30, activation="leaky_relu"),
3     Dense(30, activation="leaky_relu"),
4     Dense(1)])
5 nn.compile(optimizer="adam", loss="mse")
6 es = EarlyStopping(patience=10, restore_best_weights=True)
7 hist = nn.fit(X_train_est, y_train, epochs=100,
8               validation_data=(X_val_est, y_val), callbacks=[es], verbose=0)
```

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# Evaluating the models

MSE loss on  $\log O_k(v)$  is then a *lognormal* assumption for the outstanding, so every RMSE below is on the *log scale*, not in dollars.

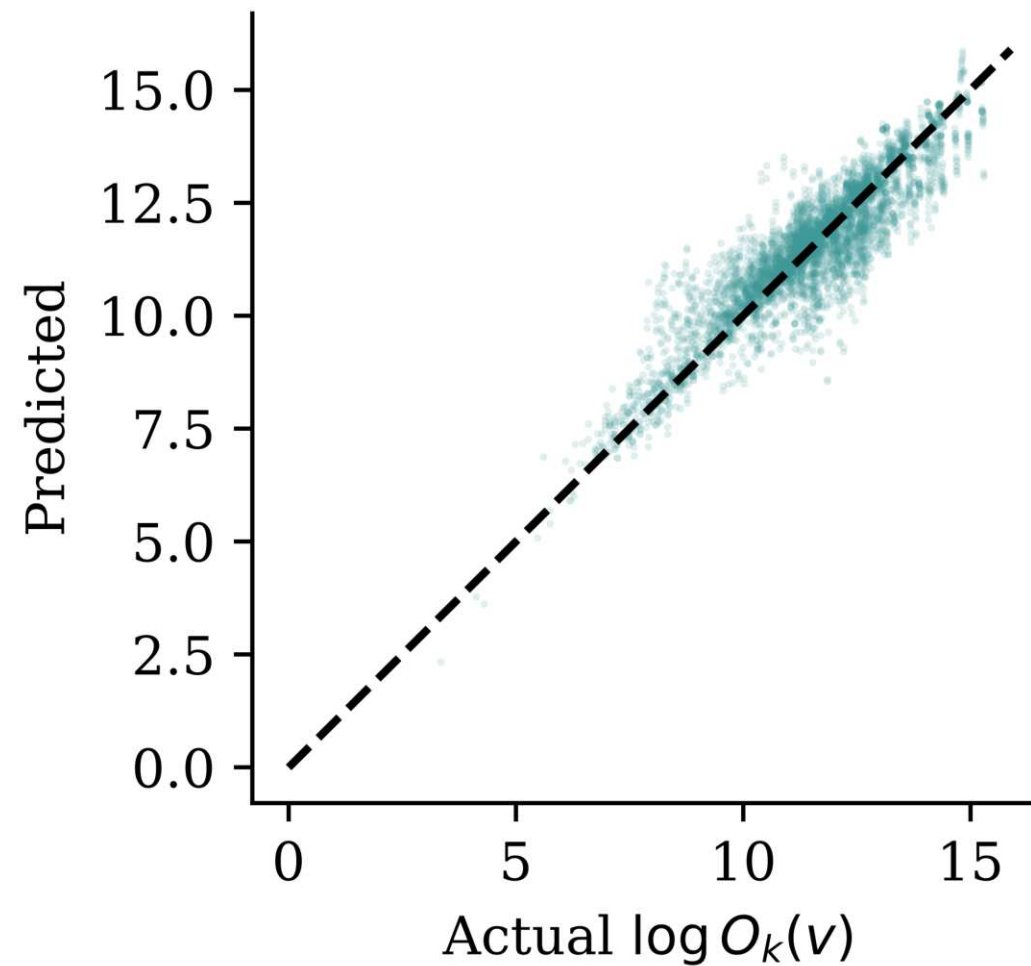
```
1 case_estimates = np.log(test["Estimate"])
2 pred_plain = glm_plain.predict(X_test_plain)
3 pred_est = glm_est.predict(X_test_est)
4 pred_nn = nn.predict(X_test_est, verbose=0).flatten()
```

|                                    |                               |
|------------------------------------|-------------------------------|
| Case estimates alone               | test RMSE (log scale) = 1.418 |
| GLM without case estimates         | test RMSE (log scale) = 1.150 |
| GLM with case estimates            | test RMSE (log scale) = 0.977 |
| Neural network with case estimates | test RMSE (log scale) = 0.693 |

## Note

On this synthetic portfolio the payment history already carries most of the signal, so the case estimates may add little as a feature. On real data, assessors know things the payment stream doesn't, and case estimates tend to matter much more.

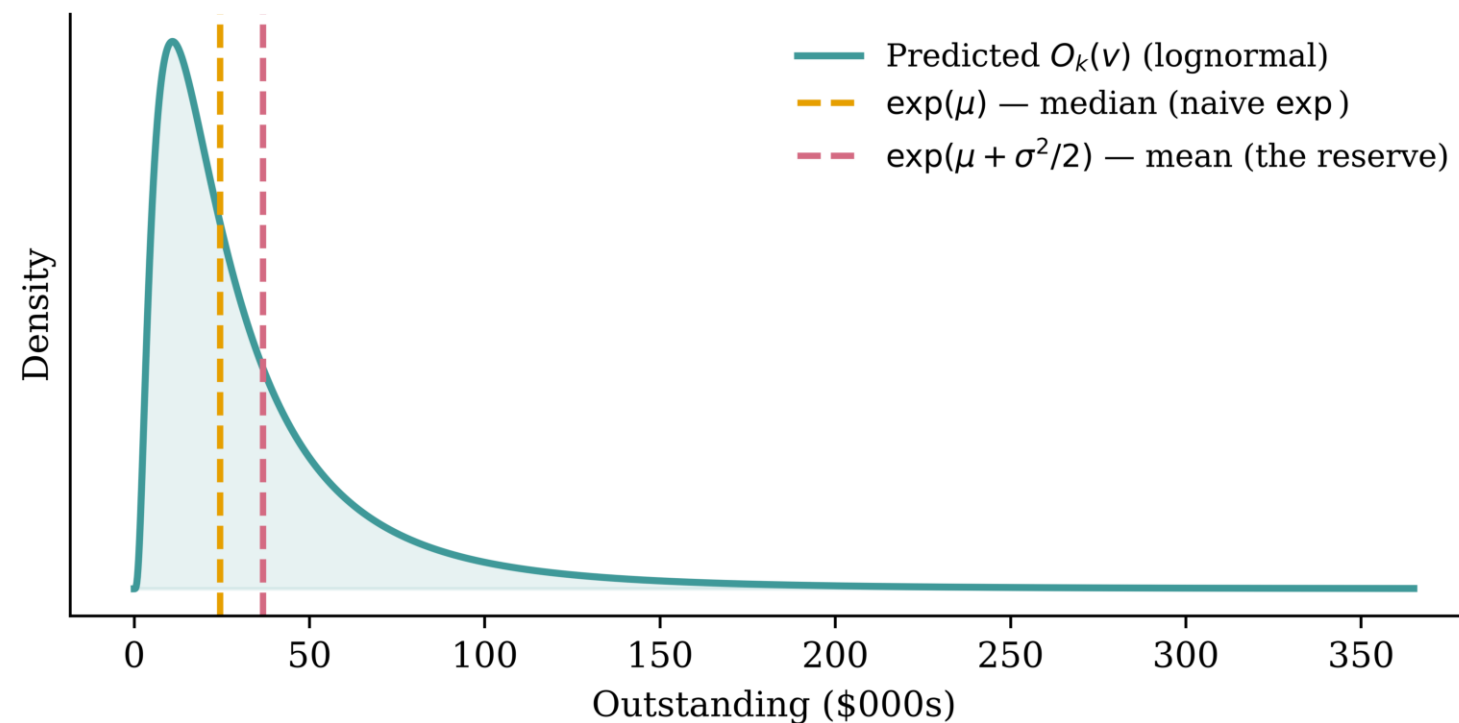
# Predicted vs actual outstanding



Each point is one test-set snapshot  $(X_{k,v}, O_k(v))$ , plotted on the model's log scale; the dashed diagonal is perfect prediction.

# Exponentiating log predictions under-reserves

Squared-error loss on  $\log O_k(v)$  is exactly the constant-variance normal assumption from distributional regression,  $\log O_k(v) \sim \mathcal{N}(\mu_k, \sigma^2)$ , so on the dollar scale  $O_k(v)$  is *lognormal*. The naive  $\exp(\mu_k)$  is its *median*; a reserve needs the *mean*,  $\exp(\mu_k + \frac{1}{2}\sigma^2)$ .



Dropping the  $\frac{1}{2}\sigma^2$  term predicts the median and *systematically under-reserves*; we add it back, estimating  $\sigma$  by the log-scale RMSE.

# Converting back to dollars

```
1 def to_dollars(pred_log, sigma2):           # lognormal mean, exp(mu + sigma^2/2)
2     return np.exp(pred_log + sigma2 / 2)
3
4 # sigma^2 per model: its squared validation RMSE on the log scale.
5 sigma2_plain = np.mean((y_val - glm_plain.predict(X_val_plain)) ** 2)
6 sigma2_est = np.mean((y_val - glm_est.predict(X_val_est)) ** 2)
7 sigma2_nn = np.mean((y_val - nn.predict(X_val_est, verbose=0).flatten()) ** 2)
```

# Did our model beat the professionals?

*Better%* is the share of test claims we win vs the case estimates.

```

1 actual = test["Outstanding"]
2 # How far each forecast is from the truth, proportionally (a log-scale gap).
3 case_gap = np.abs(np.log(test["Estimate"] / actual)) # the assessor's gap
4
5 def beats_assessor(pred_log, sigma2):
6     model_gap = np.abs(np.log(to_dollars(pred_log, sigma2) / actual))
7     return model_gap < case_gap
8
9 wins_plain = beats_assessor(pred_plain, sigma2_plain)
10 wins_est = beats_assessor(pred_est, sigma2_est)
11 wins_nn = beats_assessor(pred_nn, sigma2_nn)
12
13 print(f"GLM without case estimates      Better% = {wins_plain.mean():5.1%}")
14 print(f"GLM with case estimates         Better% = {wins_est.mean():5.1%}")
15 print(f"Neural network with case estimates Better% = {wins_nn.mean():5.1%}")

```

|                                    |                 |
|------------------------------------|-----------------|
| GLM without case estimates         | Better% = 35.7% |
| GLM with case estimates            | Better% = 34.1% |
| Neural network with case estimates | Better% = 38.8% |

# Weighted Better%

The portfolio reserve is dominated by the *big* claims, so weight each win by the claim's ultimate — winning a \$2m claim counts for more than winning a \$5k one:

$$\text{Weighted Better\%} = \sum_k w_k \mathbf{1}\{\text{model wins claim } k\}, \quad w_k = \frac{U_k}{\sum_j U_j}.$$

```
1 weight = test["claim_no"].map(ultimate)
2 weight = weight / weight.sum()
3
4 print(f"GLM without case estimates      Weighted Better% = "
5       f"{(weight * wins_plain).sum():5.1%}")
6 print(f"GLM with case estimates         Weighted Better% = "
7       f"{(weight * wins_est).sum():5.1%}")
8 print(f"Neural network with case estimates Weighted Better% = "
9       f"{(weight * wins_nn).sum():5.1%}")
```

|                                    |                          |
|------------------------------------|--------------------------|
| GLM without case estimates         | Weighted Better% = 52.7% |
| GLM with case estimates            | Weighted Better% = 52.2% |
| Neural network with case estimates | Weighted Better% = 56.4% |

Weighting by size lifts every model: they may trail the assessor on a plain *count* of claims, yet come out ahead on the large claims that actually move the reserve.

# Lecture Outline

- Introduction
- Insurance Claim Payments and Case Estimates
- Benchmark: Aggregate Reserving
- Individual Claim Modelling Choices
- Preprocessing One Claim
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- Case Study: Synthetic Claims
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- **Portfolio-level Metrics**
- Wrapping Up

# Benchmark: chain ladder

Per-snapshot accuracy is one lens — but a reserving model's real job is the portfolio total.

```

1 # The triangle uses the observed window (accident quarters 61-100), shifted to
2 # 1-40 so it reads as ten fresh accident years valued at quarter 100.
3 window ← subset(payments, occurrence_period > 60 & payment_period ≤ 100)
4 window$acc_year ← ceiling((window$occurrence_period - 60) / 4)
5 window$dev_year ← ceiling((window$payment_period - 60) / 4) - window$acc_year + 1
6
7 incr ← aggregate(payment_size ~ acc_year + dev_year, window, sum)
8 tri ← incr2cum(as.triangle(incr, origin = "acc_year",
9                           dev = "dev_year", value = "payment_size"))
10 round(tri / 1e6, 2) # cumulative payments, $m

```

|          | dev_year |      |       |       |       |       |       |       |       |
|----------|----------|------|-------|-------|-------|-------|-------|-------|-------|
| acc_year | 1        | 2    | 3     | 4     | 5     | 6     | 7     | 8     | 9     |
| 1        | 0.30     | 3.09 | 9.09  | 18.92 | 33.61 | 44.07 | 56.21 | 64.68 | 69.82 |
| 2        | 0.48     | 3.92 | 12.84 | 28.16 | 41.98 | 59.02 | 68.39 | 74.46 | 76.80 |
| 3        | 0.32     | 3.65 | 10.06 | 18.23 | 31.71 | 47.09 | 48.47 | 50.93 | NA    |
| 4        | 0.38     | 4.43 | 12.12 | 26.74 | 40.39 | 53.48 | 64.34 | NA    | NA    |
| 5        | 0.31     | 4.36 | 10.41 | 24.81 | 41.64 | 55.46 | NA    | NA    | NA    |
| 6        | 0.31     | 3.41 | 11.87 | 21.74 | 35.67 | NA    | NA    | NA    | NA    |
| 7        | 0.27     | 3.40 | 9.45  | 17.67 | NA    | NA    | NA    | NA    | NA    |
| 8        | 0.33     | 3.45 | 10.43 | NA    | NA    | NA    | NA    | NA    | NA    |
| 9        | 0.35     | 3.17 | NA    | NA    | NA    | NA    | NA    | NA    | NA    |
| 10       | 0.36     | NA   | NA    | NA    | NA    | NA    | NA    | NA    | NA    |

# The chain ladder reserve

```
1 mack ← MackChainLadder(tri, est.sigma = "Mack")
2 reserve_cl ← summary(mack)$Totals["IBNR:", ]
3 true_outstanding ← sum(subset(payments, payment_period > 100)$payment_size)
4
5 cat(sprintf("Chain ladder reserve:   $%.1fm\n", reserve_cl / 1e6),
6     sprintf("True total outstanding: $%.1fm\n", true_outstanding / 1e6))
```

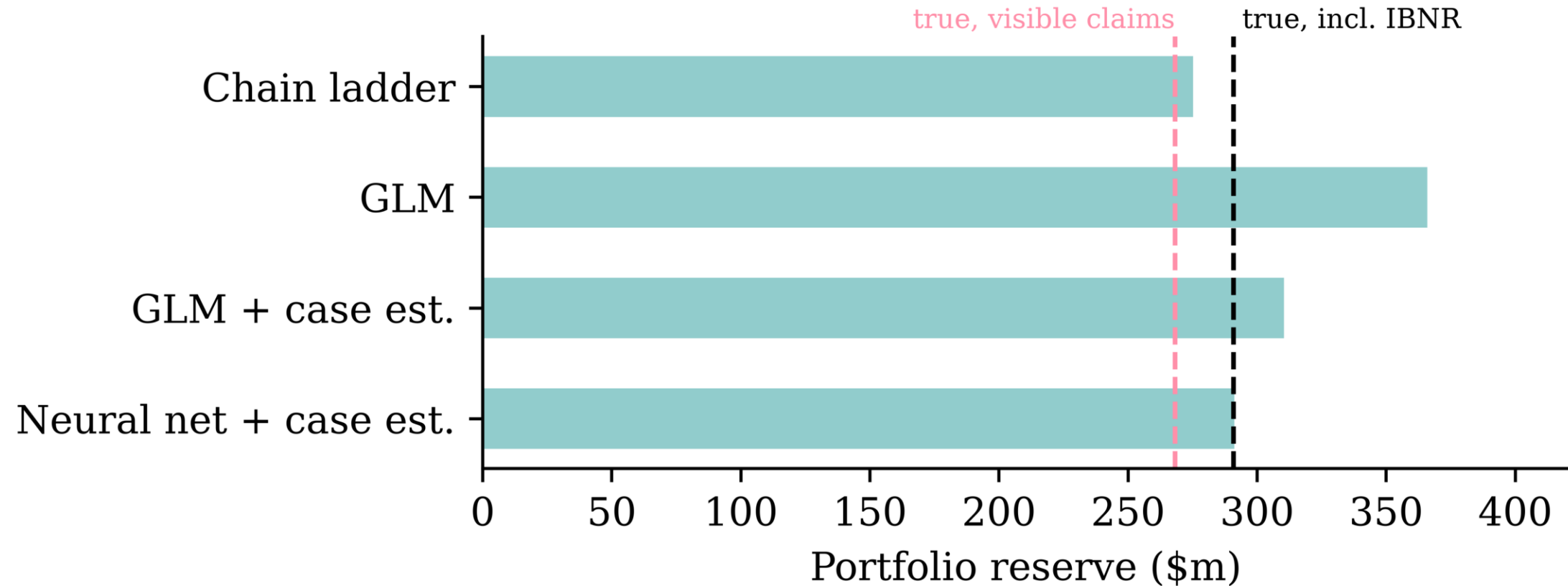
Chain ladder reserve: \$275.2m  
True total outstanding: \$290.8m

# Aggregate claim to estimate $T(v)$

Predict  $\hat{O}_k(v)$  for each still-open claim, and sum to the portfolio reserve  $\hat{T}(v)$ .

```
1 reserves = {  
2     "Chain ladder": float(r.reserve_cl),  
3     "GLM": to_dollars(glm_plain.predict(X_unfinalised_plain), sigma2_plain).sum(),  
4     "GLM + case est.": to_dollars(glm_est.predict(X_unfinalised_est), sigma2_est).sum(),  
5     "Neural net + case est.": float(to_dollars(  
6         nn.predict(X_unfinalised_est, verbose=0).flatten(), sigma2_nn).sum()),  
7 }
```

# The portfolio reserves compared



The individual models can only reserve for claims they can *see*, so their fair target is the lower value (left line); the chain ladder triangle implicitly projects the IBNR claims too, so its target is the larger value (right line).

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# Individual vs aggregate reserving

- *Aggregate* (chain ladder on triangles): simple, stable, industry standard; but throws away claim-level covariates and gives no per-claim view.
- *Individual*: predicts  $O_k(v)$  per open claim; the portfolio reserve is just the sum  $T(v)$ ; supports triage and segmentation; per-claim errors partially cancel in the aggregate.
- Individual level reserving is a data engineering headache, and the total still misses IBNR, so an individual model alone is *not yet* a full portfolio reserve.

# What we didn't cover

- *IBNR*: our model only reserves claims it can see; incurred-but-not-reported claims need a separate (usually aggregate) adjustment.
- *Uncertainty*: a reserve needs a risk margin, not just a point estimate; distributional forecasts and ensembling give prediction intervals.
- *When does granular beat aggregate?*

# Python package versions

```
1 from watermark import watermark
2 print(watermark(python=True, packages="keras,matplotlib,numpy,pandas,sklearn,torch"))
```

Python implementation: CPython

Python version : 3.14.5

IPython version : 9.15.0

keras : 3.15.0

matplotlib: 3.11.0

numpy : 2.5.1

pandas : 3.0.3

sklearn : 1.9.0

torch : 2.12.1

# R package versions

```
1 cat(sessionInfo()$R.version$version.string, "\n\n")
2 for (p in c("SPICE", "Synthetic", "ChainLadder", "reticulate", "knitr"))
3   cat(sprintf("%-12s %s\n", p, as.character(packageVersion(p))))
```

R version 4.6.0 (2026-04-24)

|             |        |
|-------------|--------|
| SPICE       | 1.1.2  |
| Synthetic   | 1.1.1  |
| ChainLadder | 0.2.21 |
| reticulate  | 1.46.0 |
| knitr       | 1.51   |

# References

- Avanzi, B., Taylor, G., & Wang, M. (2023). SPLICE: A synthetic paid loss and incurred cost experience simulator. *Annals of Actuarial Science*, 17(1), 7–35. <https://doi.org/10.1017/S1748499522000057>
- Avanzi, B., Taylor, G., Wang, M., & Wong, B. (2021). SynthETIC: An individual insurance claim simulator with feature control. *Insurance: Mathematics and Economics*, 100, 296–308. <https://doi.org/10.1016/j.insmatheco.2021.06.004>